

SUPPLEMENTARY APPENDIX

Supplementary Table A1. Study framing

Study element	Definition
Topic	Association between PM2.5 and PM2.5-related respiratory diseases in Northern Thailand
Research question	Can next-day PM2.5 prediction from previous-day information help hospitals prepare additional resources for an increase in respiratory patients?
Objective	To model the association between predicted PM2.5 and respiratory diagnosis records.
Hypothesis	Predicted PM2.5 is linearly associated with respiratory diagnosis records.
Expected use	Support the provincial working group in preparing staff, beds, supplies, and patient-flow plans for PM2.5-related respiratory disease periods.

Supplementary Table A2. Data sources, acquisition routes, and principal roles

Source	Acquisition	Principal role
OpenAQ observed PM2.5	Key API	Daily observed PM2.5 by sensor location
Open-Meteo modeled PM2.5	Air-quality API	Daily modeled PM2.5 benchmark
Open-Meteo weather	Archive API	Daily meteorological features
NASA FIRMS VIIRS	Key API	Satellite fire-hotspot records
DDC/MOPH respiratory records	Open data source	Respiratory diagnosis records
MOPH facility data	Open data source	Facility and hospital-capacity context

Supplementary Table A3. Reproducible execution pipeline

Processes	File	Output
0	requirements.txt	Environment and import gate
1	fetch_data.py	Raw bytes and acquisition manifest
2	prepare_tables.py	Eleven standardized source-aligned tables
3	prepare_data.py	AT1 and AT2 analysis-ready/model-ready tables
4	check_dataset.py	Feasibility gate; continue only when FAIL = 0
5	analysis.py	Descriptive tables and figures
6	modeling.py	AT2 estimates; AT1 predictions and metrics
7	manifests and logs	Release verification

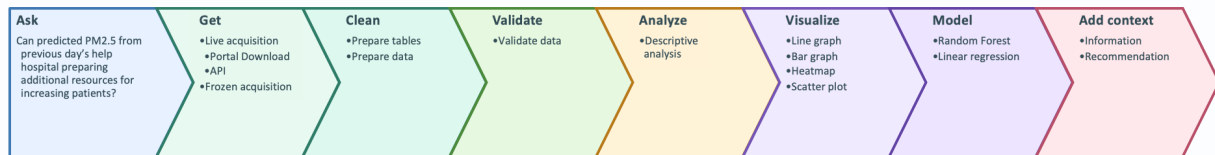
Supplementary Table A4. Primary, foreign, and analytical merge keys of the source-aligned tables

Source-aligned table	Grain	Primary key	Foreign / relational key	Analytical merge key
provinces.csv	One province	province_key	None	Reference only
location.csv	One monitoring location	location_id	province_key → provinces.csv	Not directly merged
observepm25_sensorday.csv	One sensor-day	sensor_id + analysis_date	location_id → location.csv; province_key → provinces.csv	Not directly merged
observepm25_provinceday.csv	One province-day	province_key + analysis_date	province_key → provinces.csv	province_key + analysis_date
weather_provinceday.csv	One province-day	province_key + analysis_date	province_key → provinces.csv	province_key + analysis_date
modelpm25_provinceday.csv	One province-day	province_key + analysis_date	province_key → provinces.csv	province_key + analysis_date
hotspot.csv	One province-day	province_key + analysis_date	province_key → provinces.csv	province_key + analysis_date
facility.csv	One health facility	facility_key	province_key → provinces.csv	Not directly merged
population.csv	One province-year	province_key + year_ce	province_key → provinces.csv	province_key + year_ce
hospital.csv	One health facility	facility_key	facility_key → facility.csv; province_key → provinces.csv	Aggregate, then province_key
diagnosis.csv	One province-month	province_key + year_month	province_key → provinces.csv	province_key + year_month

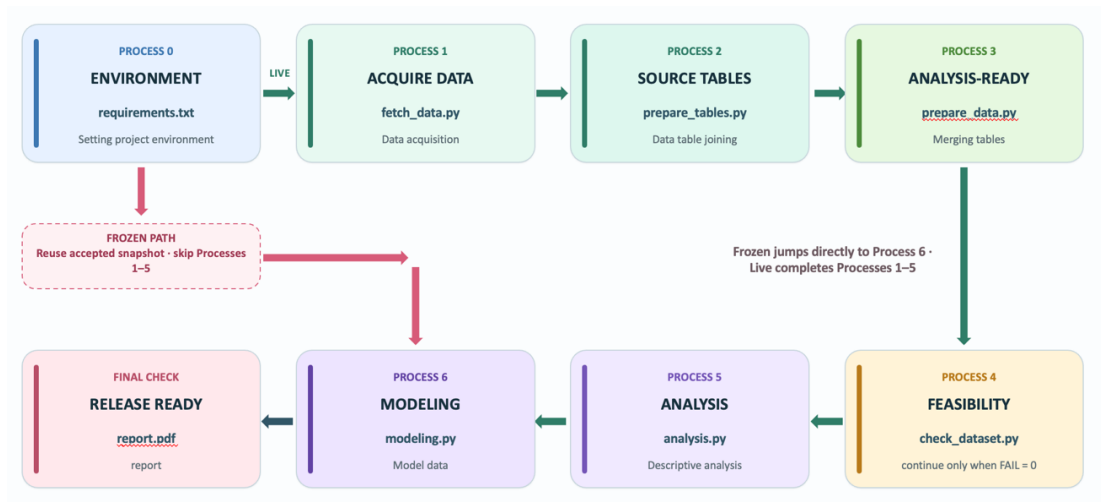
Supplementary Table A5. Core keys and analytical measures

Field / concept	Type	Definition	Unit / format	Main table or output
province_key	Exact field	Stable province identifier used across source-aligned tables	Categorical key	All province-linked tables
location_id	Exact field	Monitoring-location identifier	Categorical key	location.csv; observepm25_sensorday.csv
sensor_id	Exact field	Observed PM2.5 sensor identifier	Categorical key	observepm25_sensorday.csv
facility_key	Exact field	Stable health-facility identifier	Categorical key	facility.csv; hospital.csv
analysis_date	Exact field	Thailand-local calendar date used for daily alignment	Date (YYYY-MM-DD)	Daily tables
year_ce	Exact field	Calendar year used to join annual population	Four-digit year	population.csv
year_month	Exact field	Calendar month used for health analysis	YYYY-MM	diagnosis.csv; AT2
Observed PM2.5	Measure role	Daily sensor-based PM2.5 summarized to province-day	µg/m ³	observepm25_provinceday.csv
Modeled PM2.5	Measure role	Daily modeled PM2.5 used as an optional benchmark or feature	µg/m ³	modelpm25_provinceday.csv
Weather features	Measure set	Daily meteorological conditions used in analysis	Source-specific units	weather_provinceday.csv
Hotspot count	Measure role	Daily VIIRS hotspot count summarized by geography/distance band	Count	hotspot.csv
Population	Measure role	Annual province population used for denominators/context	Persons	population.csv
Hospital capacity	Measure set	Hospital and opened-bed context aggregated to province	Facility/bed count	hospital.csv
Respiratory diagnoses	Outcome role	Reported respiratory diagnosis count at province-month level	Count	diagnosis.csv

Field / concept	Type	Definition	Unit / format	Main table or output
AT1 prediction target	Derived field	Observed PM2.5 for the next eligible day	µg/m ³	AT1 model frame
AT1 predicted PM2.5	Model output	Model estimate of next-day observed PM2.5	µg/m ³	AT1 prediction output
AT2 exposure	Derived field	Province-month predicted PM2.5 summary used in association analysis	µg/m ³	AT2 model frame
AT2 outcome	Model outcome	Province-month respiratory diagnosis burden	Count	AT2 model frame



Supplementary Figure B1. Data science approach for the PM2.5 project



Supplementary Figure B2. Reproducible data and analysis workflow

CHECKPOINTS FROM DATA EXPLORATION • C1–C4

C1	TIME EVIDENCE OpenAQ and weather timestamps were retained. FIRMS date and zero-padded time were combined as UTC. All sources were converted to Asia/Bangkok before deriving the daily date. FLAG UTC records near midnight—and hotspots after 17:00 UTC—can belong to the next calendar day in Thailand. DECISION Derive analysis_date from Thailand-local time, then aggregate observations and create hotspot lags. Retain UTC and local timestamps for audit.	JOIN EVIDENCE The complete daily grid had 13,656 rows. Of these, 10,047 had an observed PM2.5 target and 8,222 were model-ready. The monthly grid had 288 possible rows and 35 eligible rows. Final keys were unique. FLAG Unexpected row expansion suggests duplicate keys; unexplained row loss suggests an incorrect join or filter. DECISION Assert key uniqueness before every merge and left-join sources to complete grids. Explain all later exclusions through missingness and eligibility rules.
C3	MISSING VALUES EVIDENCE Observed PM2.5 was unavailable for 3,609 of 13,656 province-days (26.43%). Missingness was checked by column. A confirmed zero hotspot was kept separate from an unavailable FIRMS interval. FLAG A field with 0% missing may be a constructed key, planned grid date or derived variable—not proof that the source observed every day. DECISION Never impute the PM2.5 target. Keep unavailable intervals missing and preserve missingness flags; apply complete-case requirements only when creating eligible analysis rows.	PLACE EVIDENCE Sensor, facility and 77-province boundary records were standardized to province_key. The eight target provinces had distinct PM2.5 series and unequal coverage. FLAG Sensors and facilities are points; boundaries are areas. None implies uniform province-wide exposure. DECISION Use province-day sensor medians and province-month active-reporter counts; do not perform nearest sensor-hospital matching. Boundaries validate assignments only.

Supplementary Figure B3. Checkpoints from data exploration (C1–C4)

CHECKPOINTS FROM MODELING AND DECISION USE • C5–C6

C5	MODEL VALIDATION EVIDENCE AT1 contained 8,222 model-ready province-days. Candidate models were compared with time-aware train, validation, and test separation. AT2 contained 35 eligible province-months. FLAG Random splitting, same-day target information, or preprocessing fitted on all data can leak future information. The small AT2 frame makes estimates sensitive to specification. DECISION Fit preprocessing on training data only; tune on validation data; lock the model before the held-out test; and report one authoritative export for each result.
C6	GROUND TRUTH & DECISION USE EVIDENCE Predicted and observed PM2.5 showed similar overall direction in held-out comparison. Health and capacity data add province-level context for preparedness. FLAG Prediction is not causal proof. Hotspot features do not identify a specific fire as the cause, and bed data do not represent real-time occupancy or staffing pressure. DECISION Use the output as an early-warning input. Confirm with current sensor, hotspot, and local capacity information before changing beds, staff, supplies, or patient-flow plans.

Supplementary Figure B4. Checkpoints from modeling and decision use (C5–C6)